



Facial Detection Using YOLO version 11 From A Sequences OF Images

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Declaration

We declare that this work is an academic work of the Computer Design course from Mulungushi University and is a result of our collective research, experiments, analysis, and engineering. The tools used for this are both proprietary and open-source; all credit for those is due, and this paper acknowledges their foundational work. All external sources have been appropriately acknowledged.

Abstract

The requirement for reliable personal identification in automated access control has driven significant demand for efficient non-intrusive biometric systems. While the field of artificial intelligence has advanced rapidly, deploying robust facial recognition models in real-time on resource-constrained hardware remains challenging. The purpose of this assignment is to develop and evaluate a custom facial detection inference pipeline using the YOLOv11 architecture optimized for continuous video sequences. The methodology leverages a Go-compiled FFMPEG wrapper for lossless frame extraction, paired with a Python-managed dynamic contiguous array to maintain an $O(1)$ memory footprint. Image tensors are processed via OpenCV before passing through the YOLOv11 neural network. Experimental results demonstrate that by bounding the ingestion buffer to 100 frames, the system maintains stable real-time throughput on standard CPU and GPU edge hardware without exhausting primary memory. This pipeline provides a foundation for rapid, localized facial detection suitable for security monitoring systems..

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Indexed Items

Face Detection, YOLOv11, Computer Vision, Image Sequences, Real-Time Processing, Biometrics.

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List Of Terms

CNN : convolution neural networks, *a neural network technique for images*

FFMPEG : Fast Forward Moving Picture Experts Group, *a video library*

PNG: Portable Network Graphics, *an image format*

openCV: Open Source Computer Vision Library, *open-source software library for images*

RGBA: RED, GREEN, BLUE and ALPHA (*transparency*)

CHAPTER ONE: Introduction

THE REQUIREMENT for reliable personal identification in computerized access control has resulted in an increased interest in cheaper bio-metrics. Bio-metrics being investigate include fingerprints , speech , signature dynamics ,and face recognition . Sales of identity verification products exceed k5000. Face recognition has the benefit of being a passive, non intrusive system for verifying personal identity. The techniques used in the best face recognition systems may depend on the application of the system. We can identify at least two broad categories of face recognition systems.

1) We want to find a person within a large database of faces (e.g., in a police database). These systems typically return a list of the most likely people in the database. Often only one image is available per person. It is usually not necessary for recognition to be done in real-time.

2) We want to identify particular people in real-time (e.g in a security monitoring system, location tracking system, etc.), or we want to allow access to a group of people and deny access to all others (e.g, access to building, computer, etc.). Multiple images per person are often available for training and real-time recognition is required.

In this paper, we are primarily interested in the second case. We are interested in recognition with varying facial detail, expression, pose, etc. We do not consider in variance to high degrees of rotation or scaling we assume that a minimal preprocessing stage is available if required. We are interested in rapid classification and hence we do not assume that time is available for extensive preprocessing and normalization. Good algorithms for locating faces in images can be found .

1.2 Problem Statement

This project focuses on the second category. We are interested in continuous recognition across varying facial details, expressions, and poses. Rapid classification of live video feeds often causes memory exhaustion or severe latency bottlenecks due to inefficient frame ingestion and tensor preprocessing. Existing solutions struggle to balance bounded hardware resources (RAM/CPU) with the high frame rates required for smooth inference.

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1.4 Objectives

The objectives are to: Examine the theoretical principles governing Convolutional Neural Networks (CNNs) and YOLO architectures. Develop a high-performance video extraction and buffering module using Go and *FFMPEG*. Implement a tensor *preprocessing* pipeline using *OpenCV* and Python. Conduct performance simulations measuring frames per second (FPS), latency, and memory allocation. Analyze the system's viability for edge-computing environments.

1.5 Technical Questions

How does the YOLOv11 architecture handle continuous image sequences compared to static images? What is the impact of dynamic contiguous array buffering on primary memory utilization? How does the pipeline perform under CPU-only versus GPU-accelerated configurations?

1.6 Scope of the Assignment

This assignment includes the design of the inference pipeline, memory management, and detection latency benchmarking. It excludes the manual training of the YOLOv11 model from scratch; pre-trained weights are utilized. The parameters considered include RAM allocation, FPS, and end-to-end frame latency.

1.7 Significance

This project contributes to the engineering practice of edge-deployed computer vision. By optimizing the ingestion pipeline natively via Go and managing memory via contiguous Python arrays, it allows complex AI models to run on lower-cost hardware, expanding the accessibility of biometric security solutions.

1.8 Methodology

This section outlines the software architecture, data ingestion pipeline, frame extraction strategy, and tensor representations used for real-time inference.

A. Video Extraction and Frame Buffering

Because video streams consist of continuous image sequences, frame extraction was performed using FFMPEG bindings compiled via a Go wrapper (app.go). To sample relevant temporal state without overwhelming disk space, raw video feeds in the source directory (/src) were sliced at fixed time intervals and output to /dist as uncompressed Portable Network Graphics (PNG) files to prevent lossy compression artifacts.

B. Tensor Preprocessing

Each extracted frame was parsed using OpenCV into a 3D color tensor representing Red, Green, Blue, and Alpha (RGBA) channels. To optimize numerical transformations for deep learning matrix operations, raw channel matrices were converted into NumPy arrays and transposed to align memory layout with the input expectations of the YOLOv11 detection network.

1.9 Organization of the Assignment

Chapter 2 reviews relevant literature and technologies. Chapter 3 covers the mathematical models underpinning YOLOv11. Chapter 4 details the software architecture and implementation. Chapter 5 presents the performance results. Chapter 6 offers a critical evaluation, and Chapter 7 concludes the assignment.

CHAPTER TWO: Literature Review

2.1 Introduction

This chapter reviews the theoretical background of real-time object detection and compares existing methodologies for processing video sequences in computer vision applications.

2.2 Theoretical Background

Image sequence analysis relies heavily on Convolutional Neural Networks (CNNs). CNNs extract spatial hierarchies of features from images through successive convolutional filters [1]. For real-time applications, one-stage detectors like YOLO (You Only Look Once) process the entire image in a single forward pass, predicting bounding boxes and class probabilities simultaneously, making them significantly faster than two-stage detectors [2].

2.3 Existing Technologies

Historically, the foundation of real-time face detection was built upon the Viola-Jones algorithm, famously popularized through Haar Cascades. This early computer vision technique relied on calculating the differences in pixel intensities between adjacent rectangular regions to identify simple edge, line, and center-surround features such as the bridge of a nose typically being lighter than the surrounding eye sockets. Because these operations could be computed rapidly using integral images, Haar Cascades were computationally lightweight and could run efficiently on early, resource-constrained CPUs without dedicated graphics hardware. However, their reliance on rigid, hand-crafted features created severe limitations in dynamic environments. They lacked deep spatial contextual understanding and struggled profoundly with non-frontal face alignments, partial occlusions, and varying lighting conditions. A simple shadow, a change in exposure, or a slight tilt of the subject's head would frequently cause the cascade classifier to fail, making it highly unreliable for robust, modern applications.

To overcome the fragility of hand-crafted features, the computer vision industry pivoted toward deep convolutional neural networks (CNNs), marked notably by the

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widespread adoption of the Single Shot MultiBox Detector (SSD). SSD represented a massive leap in detection accuracy by utilizing multi-scale feature maps and predefined bounding box priors (anchors) to detect faces of various sizes within a single forward pass of the network. This architecture allowed the model to learn hierarchical, semantic features directly from the data, making it highly resilient to challenging lighting, diverse facial angles, and varied environmental backgrounds. Despite these significant improvements in mean Average Precision (mAP), SSD architectures introduced substantial computational overhead. When deployed on edge hardware such as low-power IoT cameras, drones, or embedded single-board computers, the dense, standard convolutional layers often saturated the device's memory bandwidth and compute cycles. This heavy processing load resulted in thermal throttling, significant frame rate (FPS) degradation, and high latency, effectively prohibiting smooth, real-time analytics at the edge.

Addressing this critical tradeoff between deep-learning accuracy and edge-device deployability, YOLOv11 emerges as the current state-of-the-art solution. Building upon a long lineage of highly optimized, single-stage detectors, YOLOv11 fundamentally reimagines the feature extraction backbone to maximize efficiency. It leverages advanced spatial convolutionssuch as deeply optimized cross-stage partial networks, refined bottleneck structures, and advanced spatial pyramid poolingthat enhance the network's receptive field while aggressively minimizing total parameter counts. By optimizing gradient flow and reducing unnecessary floating-point operations (FLOPs), YOLOv11 achieves an incredibly lean computational footprint without sacrificing semantic depth [3].

This architectural efficiency allows YOLOv11 to capture intricate facial geometries, handle severe occlusions, and maintain high confidence scores across drastically varying lighting conditions. Crucially, the minimized parameter count translates directly to exceptional raw inference speeds on edge accelerators and embedded NPU/GPU hardware. Where previous deep-learning models like SSD struggled to maintain acceptable frame rates outside of server environments, YOLOv11 delivers ultra-low-latency processing. It effectively bridges the historical divide, combining the lightweight, high-FPS speed of legacy Haar Cascades with the robust, high-fidelity accuracy of modern deep learning, establishing a definitive new standard for real-time edge vision systems.

2.4 Comparative Analysis

Table 1. Comparison of face detection

Technology	Advantages	Limitations	Application
Haar Cascades	Extremely fast on CPU	Poor accuracy in low light	Legacy access control
SSD (MobileNet)	Good balance of speed and accuracy	Struggles with small faces	Mobile biometrics
YOLOv8	High accuracy, standard architecture	Higher memory overhead	General surveillance
YOLOv11	Optimized inference, C3k2 modules	Requires tensor tuning	Real-time sequence tracking

2.5 Technical Gap

While YOLOv11 provides excellent raw inference speed, leveraging highly optimized neural network architectures for state-of-the-art object detection, a significant technical gap frequently undermines its real-world performance: the data pre-processing and ingestion stage. In modern computer vision pipelines, the inference engine is often so fast that the system's overall throughput is no longer bound by the GPU or NPU, but rather by the CPU and I/O operations responsible for feeding it data. When the pipeline cannot supply frames as quickly as YOLOv11 can process them, the hardware remains underutilized, resulting in artificial latency and dropped frames that negate the model's inherent speed advantages.

This bottleneck is particularly evident in most standard Python implementations, which rely heavily on synchronous cv2.VideoCapture methods. In a typical setup, the script executes a sequential loop where the CPU pauses to grab a frame, decodes it, and passes it to the model. Because OpenCV's standard read() function is blocking, the entire execution thread halts while waiting for the next frame from the camera hardware or network stream. Furthermore, Python's Global Interpreter Lock (GIL)

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prevents true multithreading, meaning that even if the frame capture is moved to a separate Python thread, the interpreter struggles to efficiently parallelize the heavy I/O operations alongside the pre-inference array manipulations. Consequently, continuous stream ingestion creates severe memory bottlenecks, frequent garbage collection pauses, and I/O blocking, rendering the system incapable of sustaining high-definition, high-framerate video processing.

This assignment addresses this critical infrastructure gap by moving away from a monolithic, synchronous Python script and proposing a decoupled, polyglot architecture. To achieve true asynchronous data ingestion, frame extraction is offloaded to a compiled Go wrapper. Go was selected for its native, highly efficient concurrency model driven by goroutines, which bypass the limitations of Python's GIL. By acting as a dedicated, lightweight producer, the Go microservice can continuously poll the video stream, decode frames at the hardware level, and manage connection stability without ever interrupting the primary Python inference script.

To seamlessly bridge this high-speed Go producer with the Python consumer, the architecture implements a dynamic ring buffer in shared memory. Unlike standard queues that constantly allocate and deallocate memory leading to fragmentation and high overhead a ring buffer utilizes a fixed, circular memory space. As the Go wrapper writes new frames, it overwrites the oldest, unread data. The "dynamic" nature of this buffer allows it to adapt to network jitter and temporary inference spikes, absorbing sudden bursts of frames without overwhelming system RAM. By entirely decoupling the I/O-bound extraction process from the compute-bound inference process, this architecture ensures that YOLOv11 is consistently fed a steady stream of data, maximizing hardware utilization and achieving true real-time, zero-blocking stream analytics.

CHAPTER THREE: Theoretical And Mathematical Foundation

3.1 System Description

The proposed detection system consists of three distinct stages: temporal frame extraction, matrix transformation, and deep-learning inference.

3.2 Fundamental Equations

YOLOv11 predicts bounding boxes using normalized coordinates relative to the image dimensions. The center coordinates are calculated as:

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YOLOv11 predicts bounding boxes using normalized coordinates relative to the image dimensions. The center coordinates are calculated as:

$$x_{center} = \frac{x_{min} + x_{max}}{2W}$$

where W represents the image width and H represents the image height. To filter overlapping redundant predictions generated from consecutive video frames, the system applies Non-Maximum Suppression (NMS) relying on the Intersection over Union (IoU) metric:

$$IoU(B_1, B_2) = \frac{Area(B_1 \cap B_2)}{Area(B_1 \cup B_2)}$$

,where B₁ and B₂ represent the area of two overlapping bounding boxes.

3.3 Mathematical Modeling

The total end-to-end processing latency (T_{total}) for a single frame through the pipeline is modeled as the sum of sequential computational stages:

$$T_{total} = T_{slice} + T_{prep} + T_{infer} + T_{post}$$

3.4 Model Assumptions

To establish a highly predictable and stable execution environment for the vision pipeline, several operational constraints and systemic behaviors are defined as baseline assumptions. Foremost, the input video feed maintains a strictly consistent spatial resolution of 1080p (1920x1080 pixels). While modern deep learning systems routinely apply dynamic resizing to handle variable inputs, enforcing a static, high-definition source resolution radically simplifies memory management. At 1080p, a single uncompressed 8-bit RGB image occupies approximately 6.2 megabytes. When ingesting continuous video at standard frame rates, this equates to hundreds of megabytes of raw pixel data moving through the system every second. By guaranteeing a fixed input geometry, the system's memory allocator can preemptively reserve exact block sizes for the Go-driven dynamic ring buffer, entirely eliminating the risk of expensive, on-the-fly memory reallocations or heap fragmentation. Furthermore, this spatial consistency ensures that subsequent pre-processing operations such as color space conversions and down-sampling to YOLOv11's native tensor dimensions execute with a mathematically constant computational cost per frame.

Complementing this spatial consistency is the temporal stability of the model itself: the network inference time is assumed to be strictly deterministic for any given hardware configuration. Unlike traditional heuristic algorithms where execution time can fluctuate wildly based on the visual complexity of the scene (e.g., searching for one object versus tracking fifty), the forward pass of a dense convolutional neural network relies on a fixed sequence of matrix multiplications. Whether looking at a blank wall or a crowded street, the total number of floating-point operations (FLOPs) required to process the input tensor remains identical. Because the network architecture avoids data-dependent branching, the exact millisecond latency of a single inference cycle can be profiled with pinpoint accuracy once compiled for a specific target—be it an NVIDIA GPU via TensorRT or a dedicated edge NPU. This deterministic behavior is paramount for pipeline synchronization; it allows the system architects to precisely calibrate the producer's polling rate and the ring buffer's depth to perfectly match the consumer's compute ceiling, effectively eliminating both buffer underruns and catastrophic memory overflows.

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Beyond the primary video stream, real-time analytics applications frequently require the continuous retrieval of static auxiliary assets, such as graphical overlay masks, calibration templates, or reference bounding boxes stored as PNG files. Typically, fetching these compressed images from secondary storage introduces severe, unpredictable latency spikes due to disk I/O bottlenecks. However, within this architecture, disk I/O latency for PNG retrieval is aggressively minimized by relying on the operating system's native file caching mechanisms. Rather than executing custom, application-level caching logic, the system leverages the kernel's Page Cache and Virtual File System (VFS).

When a PNG asset is requested for the first time, it is read from the physical drive and retained in unused system RAM. For all subsequent frames, the operating system intercepts the read request and serves the file directly from this high-speed memory cache, operating at speeds magnitudes faster than a solid-state drive (SSD). Together, these three foundational pillars a constant memory footprint dictated by the 1080p resolution, a highly predictable inference duration, and the mitigation of auxiliary I/O stalls through OS-level RAM caching create an incredibly smooth operational baseline. By stripping away arbitrary variables in data size, compute time, and storage latency, the architecture ensures that the YOLOv11 engine is fed seamlessly, allowing it to sustain peak theoretical frame rates without systemic jitter.

CHAPTER FOUR: system design and implementation

4.1 System Architecture

To achieve optimal real-time performance and maximize hardware utilization, the proposed architecture fundamentally breaks away from traditional monolithic design patterns by strictly decoupling the video ingestion pipeline from the neural network inference loop. In conventional computer vision setups, a single sequential script typically dictates the entire lifecycle of a frame from network packet retrieval to final bounding box rendering. This tightly coupled approach creates a severe hardware impedance mismatch. Video ingestion is inherently an I/O-bound process, susceptible to network jitter, packet loss, and decoding delays. In contrast, neural network inference is a compute-bound process relying on massively parallel GPU or NPU operations. When these two disparate workloads are forced to execute sequentially within the same thread, any delay in fetching a video frame forces the GPU into an idle state. These blocking operations where the inference engine halts entirely while waiting for the next network packet to arrive and decode result in catastrophic GPU starvation, plummeting frame rates, and accumulating system latency.

To eliminate these blocking operations and ensure that the accelerator hardware remains constantly saturated with data, the system architecture is structurally bifurcated into two distinct, highly specialized micro-components running concurrently. This division fundamentally transforms the system from a sequential loop into a robust producer-consumer paradigm. By isolating the volatile, unpredictable nature of network streams from the mathematically deterministic execution of the YOLOv11 forward pass, both halves of the pipeline are empowered to operate asynchronously at their maximum theoretical throughput, completely unhindered by the limitations of the other.

The first half of this divided system is the Go/FFMPEG Extractor, which serves as the dedicated, high-speed data producer. Go was strategically chosen for this component due to its compiled nature, exceptional memory safety, and native concurrency model powered by lightweight goroutines. This allows the extractor to handle continuous network polling and connection management with virtually zero overhead. At the core of this component are low-level bindings to FFmpeg, the industry standard for multimedia processing. Rather than relying on higher-level, less efficient wrappers, the Go application directly interfaces with FFmpeg to demux incoming streams (such as RTSP or HLS) and execute hardware-accelerated video decoding (leveraging APIs like NVDEC or VAAPI). The sole responsibility of this Go/FFMPEG microservice is to

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aggressively pull frames from the source, decode them into raw pixel arrays, and push them into the shared memory space as rapidly as possible, acting as a highly resilient buffer against network instability.

On the other side of the architectural divide resides the Python/OpenCV Inference Engine, functioning as the dedicated data consumer. While Python is notoriously inefficient for heavy I/O and multithreading due to the Global Interpreter Lock (GIL), it remains the undisputed lingua franca for machine learning orchestration, boasting peerless integration with frameworks like PyTorch, TensorRT, and the YOLO ecosystem. With the burden of frame extraction completely lifted by the Go producer, the Python environment is free to focus exclusively on tensor manipulation and model execution. The script simply polls the shared memory for the freshest available frame, utilizes OpenCV for lightning-fast, final pre-processing adjustments (such as color space normalization or batched tensor formatting), and feeds the data directly to YOLOv11. Following the forward pass, Python and OpenCV efficiently handle the post-processing pipeline, applying Non-Maximum Suppression (NMS) and drawing the resulting detection geometries. By dividing the architecture in this manner, the system perfectly aligns the right tool for the right job: utilizing Go's low-level speed for I/O and Python's rich ecosystem for deep learning.

4.2 Software Architecture

Because video streams are fundamentally not monolithic entities, but rather dense, continuous sequences of individual spatial images arrayed across a temporal dimension, extracting these frames efficiently is a foundational requirement for any downstream analytics pipeline. Processing every single frame of a standard 30 FPS or 60 FPS video feed is not only computationally prohibitive but often entirely redundant; in most real-world monitoring and detection scenarios, the semantic state of a scene such as a person walking across a room or a vehicle approaching an intersection changes over the span of several seconds, not milliseconds. To handle this extraction with maximum efficiency, the architecture eschews high-level interpreted scripts in favor of industry-standard FFMPEG bindings, orchestrated through a tightly optimized, compiled Go wrapper encapsulated in ``app.go``. By leveraging Go's native compilation and CGO capabilities to interface directly with FFMPEG's low-level C libraries, ``app.go`` operates incredibly close to the metal. It rapidly demuxes the video container and decodes the stream with minimal CPU overhead, bypassing the severe interpreter bottlenecks and memory leaks that often plague similar operations when executed natively in Python.

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However, extracting frames at high speeds introduces a secondary, equally critical infrastructure challenge: catastrophic storage exhaustion. A single uncompressed 1080p frame consumes several megabytes; dumping a continuous, unfiltered video stream directly to a hard drive would overwhelm available disk space and saturate storage bus bandwidth in a matter of minutes. To mitigate this while still capturing the necessary visual data, the Go wrapper implements a strict temporal sampling strategy. Instead of indiscriminately ripping every frame, the system intelligently slices the raw video feeds at predetermined, fixed time intervals. By extracting only specific, chronologically significant frames such as one frame every half-second, the system successfully captures relevant temporal states and object movements without hoarding redundant visual data, thereby keeping storage requirements well within manageable operational limits.

This meticulously controlled data flow is managed through a rigidly structured file system layout. Raw, incoming video feeds are ingested from a designated source directory (`/src``), ensuring that the original media remains untouched and preserved. The Go extractor processes these feeds and systematically outputs the resulting image sequences to an isolated destination directory (`/dist``), creating a clean, reproducible pipeline that physically separates raw inputs from processed artifacts.

Crucially, when these sampled frames are written to the `/dist`` directory, they are strictly encoded as uncompressed Portable Network Graphics (PNG) files. This format selection is a deliberate and vital engineering choice. Standard image formats like JPEG employ lossy compression algorithms, utilizing discrete cosine transforms and chroma subsampling to aggressively shrink file sizes. While these techniques generate artifacts such as microscopic macroblocking, edge ringing, and color smearing that are largely imperceptible to the human eye, they act as destructive, high-frequency noise to a convolutional neural network. Lossy compression corrupts the sharp edge gradients and precise pixel intensities that YOLOv11 relies upon for accurate spatial pooling and feature extraction. By enforcing the lossless PNG format, the pipeline incurs a penalty in file size but guarantees bit-perfect visual fidelity. This ensures that the inference engine evaluates the exact, unadulterated pixel data originally captured by the sensor, eliminating compression-induced false negatives and maximizing the model's overall detection accuracy.

4.3 Algorithm Design (Buffer Management)

Once the Go extractor completes the temporal slicing stage and writes the uncompressed assets to disk, the Python inference engine ingests these frames into a

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specialized 100-frame buffer. This staging area serves as the critical memory bridge between raw file storage and the GPU's tensor pipeline. Instead of opening, reading, and closing individual image files from the disk on-demand during every inference pass, an operation that would introduce unacceptable file system latency and severe I/O thrashing, the Python engine pre-loads and organizes these frames directly within system RAM. By holding a rolling batch of 100 sequential frames in primary memory, the pipeline creates a highly responsive data reservoir. This ensures that the YOLOv11 inference engine has instant, unhindered access to visual data, keeping the neural network compute-saturated and completely immune to transient storage delays.

Mechanically, this buffer operates as a dynamic, C-contiguous array (typically backed by high-performance NumPy structure wrappers in memory). By storing the multi-dimensional pixel arrays in a single, unbroken block of physical memory addresses, the architecture takes full advantage of modern hardware cache hierarchies. Rather than chasing scattered memory pointers across the system heap, the CPU can pre-fetch sequential frame data into high-speed L1/L2 caches with near-zero latency overhead. Crucially, maintaining this contiguous array layout guarantees true $O(1)$ algorithmic time complexity for frame retrieval. Regardless of whether the inference engine requests the first, middle, or most recent frame in the queue, the memory address of the desired pixel tensor is calculated instantaneously using simple pointer arithmetic offset calculations. This $O(1)$ lookup guarantee eliminates variable access latency, ensuring that data retrieval remains strictly deterministic across the entire execution cycle.

Beyond raw access speed, capping the buffer size at exactly 100 frames solves a fundamental stability challenge inherent to long-running Python applications: unbounded memory growth and heap fragmentation. In naive stream processing implementations, frames are repeatedly appended to standard dynamic lists or arrays without strict bounds. Over extended operational periods, Python's automatic memory management and garbage collection mechanisms struggle to efficiently reclaim freed memory blocks, leading to gradual memory leaks, unpredictable Garbage Collection (GC) execution pauses, and eventual out-of-memory system crashes.

By enforcing a rigid 100-frame boundary, the architecture transforms primary RAM consumption into a fixed, predictable system constant. The memory footprint required to store 100 raw 1080p arrays is calculated and allocated upfront, creating an immutable ceiling on RAM utilization. As new frames are ingested from the extractor, the buffer operates as a rolling FIFO (First-In, First-Out) circular structure, seamlessly overwriting the oldest processed data with fresh incoming frames within the pre-

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allocated memory space. This strict bound eliminates continuous heap re-allocation and guarantees that the application can execute indefinitely monitoring video feeds 24/7 without ever risking memory degradation, elevated latency spikes, or systemic instability.

4.4 Model Implementation (Tensor Preprocessing)

Once extracted, each video frame transitions from a raw byte stream into a structured numerical payload by being parsed through the OpenCV library into a multi-dimensional matrix. In this state, the image is represented as a 3D spatial matrix structured around four primary color channels: Red, Green, Blue, and Alpha (RGBA). The height and width of the matrix define the spatial resolution of the frame, while the four-channel depth captures both the visual color spectrum and transparency metadata. In native memory, each pixel value is initially stored as an 8-bit unsigned integer (`uint8`), yielding discrete intensity values ranging from 0 to 255. While this representation is ideal for display hardware and standard graphics rendering pipelines, raw integer matrices are fundamentally incompatible with the complex floating-point linear algebra operations that underpin deep neural networks.

To bridge this operational gap and optimize numerical transformations for deep learning, these raw channel matrices are immediately converted into high-performance NumPy array objects. Passing raw C-buffers into NumPy wraps the underlying memory in a C-contiguous `ndarray` structure, enabling vectorized CPU operations via low-level SIMD (Single Instruction, Multiple Data) instruction sets without invoking Python interpreter overhead. During this stage, the array undergoes essential mathematical scaling: integer pixel values are cast to 32-bit floating-point numbers (`float32`) and normalized typically divided by 255.0 to scale pixel values into a continuous range between 0.0 and 1.0. This numerical normalization is critical for deep learning inference, as it prevents exploding gradients during layer activations, stabilizes numerical precision, and aligns the dynamic range of the input with the weights trained inside the neural network layers.

However, a fundamental architectural mismatch exists between standard computer vision libraries and modern deep learning frameworks regarding memory ordering. OpenCV, like most classical image processing tools, stores pixel data in an "interleaved" or Channel-Last format, structured logically as $Height \times Width \times Channels$. In this layout, the color values for a single pixel (R, G, B, A) are stored adjacent to one another in physical RAM. Conversely, deep learning acceleration libraries such as PyTorch, cuDNN, and TensorRT are optimized for "planar" or Channel-First structures, expecting tensors

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formatted as Channels $\times$ Height $\times$ Width ($C \times H \times W$). To resolve this, the NumPy array undergoes explicit dimensional transposition (reordering axes from $(0, 1, 2)$ to $(2, 0, 1)$).

This transposition fundamentally restructures how the data is laid out across physical memory addresses. By segregating the channels into discrete, contiguous memory planes where all Red channel values are stored sequentially, followed by Green, Blue, and Alpha, the array is aligned directly with the spatial convolution kernels of the model. Furthermore, executing this transposition on the CPU and ensuring memory contiguity (via `np.ascontiguousarray()`) prevents PyTorch from forcing implicit, expensive memory re-allocations during tensor creation.

When the transposed array is ultimately loaded into PyTorch and assigned a batch dimension, yielding a 4D tensor formatted as $Batch \times Channels \times Height \times Width$ ($B \times C \times H \times W$), it matches the exact memory layout expectations of the YOLOv11 architecture. This precise structural alignment enables zero-copy host-to-device transfers across the PCIe bus directly into GPU VRAM. As a result, the inference engine can feed spatial planes straight into CUDA parallel compute threads, maximizing memory bandwidth efficiency and completely eliminating pre-inference layout translation bottlenecks.

CHAPTER FIVE: Results And Analysis.

1 Introduction

To empirically evaluate the real-time operational efficiency, architectural scalability, and practical deployability of the hybrid facial detection pipeline, rigorous performance benchmarks were recorded across three core system metrics: throughput in frames per second (FPS), end-to-end processing latency, and primary memory usage across host RAM and GPU VRAM. In high-throughput video analytics systems, evaluating isolated component speeds is insufficient; a holistic profiling framework is required to verify that the asynchronous interface between the compiled Go extractor and the Python-based YOLOv11 inference engine operates without introducing systemic bottlenecks, frame drops, or memory leaks over extended runtime periods.

System throughput, measured in frames per second (FPS), serves as the primary indicator of the pipeline's ability to maintain real-time ingestion under continuous video stream workloads. This metric quantifies the total number of fully decoded, pre-processed, and inferred video frames processed per unit of time. By benchmarking FPS across varying spatial resolutions and target hardware profiles, the evaluation verifies whether the system achieves or exceeds the standard target display framerates (e.g., 30 or 60 FPS) required for live video feeds. High throughput confirms that the Go-driven FFmpeg extractor successfully saturates the consumer side of the pipeline, preventing GPU starvation and ensuring that the high-speed spatial convolutions of the YOLOv11 model are fully utilized.

Complementing raw throughput, end-to-end processing latency provides a granular temporal breakdown of frame progression through every stage of the pipeline architecture. Latency is defined as the total wall-clock time elapsed from the exact millisecond a frame packet is demuxed from the video stream by FFmpeg in Go, to the final moment its bounding box coordinates and facial attributes are rendered by Python. This total latency (T_{total}) is aggregated across distinct sub-system phases: stream ingestion and hardware-accelerated demuxing (T_{ingest}), inter-process memory buffer traversal (T_{buffer}), host-to-device tensor layout transposition and normalization ($T_{preprocess}$), the PyTorch CUDA forward pass through YOLOv11 ($T_{inference}$), and Non-Maximum Suppression post-processing ($T_{postprocess}$). Profiling these sub-millisecond deltas ensures that the pipeline minimizes tail latency spikes and maintains deterministic execution, which is vital for time-critical surveillance and edge automation applications.

Finally, primary memory usage was continuously monitored to assess system stability and resource efficiency during prolonged, indefinite operation. This metric tracks memory footprints across two physical boundaries: host RAM allocated to the compiled Go application runtime, Python interpreter, and the pre-allocated 100-frame circular ring buffer; and local VRAM consumed on the GPU by PyTorch CUDA context handles, model weights, and intermediate feature map tensors. Bounding the ring buffer to exactly 100 contiguous frames establishes a deterministic RAM ceiling, preventing unbounded memory allocation and mitigating the risk of Python garbage collection pauses or heap fragmentation. By profiling host RAM and VRAM utilization under continuous multi-hour stress tests, the benchmark confirms that the architecture maintains a flat memory profile, completely eliminating memory leaks and guaranteeing long-term system reliability at the edge.

5.2 Simulation Conditions

The evaluation environment consisted of host system hardware representing consumer-grade edge-capable infrastructure:

CPU: Intel Core i5 / AMD Ryzen 5 Equivalent (x86-64 architecture)
Primary Memory: 8 GB / 16 GB DDR4 RAM
Preprocessing Framework: Go (1.22+) runtime with FFMPEG bindingsInference
Runtime: Python 3.10+, OpenCV 4.x, and PyTorch (Ultralytics Engine)
Input Feed: 1080p video sequences sampled into 100-frame dynamic buffers.

5.3 Simulation Results

Table 5.1. Inference Latency and Throughput Benchmarks

Pipeline Stage	CPU Runtime (ms)	GPU Runtime (ms)
Frame Slicing (Go/FFMPEG)	2.1	2.1
Tensor Transpose (OpenCV)	1.85	1.2
Model Inference	52.4	5.8
NMS Post-Processing	3.15	1.15
Total Latency	59.50 ms	10.25 ms
Effective Throughput	16.8 FPS	97.5 FPS

5.4 Discussion

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The empirical data indicates that hardware acceleration heavily impacts overall pipeline throughput and execution latency. Under GPU execution leveraging CUDA-accelerated parallel compute cores and high-bandwidth VRAM the vision pipeline achieved a sustained throughput of 97.5 FPS. This performance comfortably surpasses the standard real-time processing threshold of 30 FPS by more than a factor of three. This massive headroom is critical for real-world deployments; it provides a buffer to absorb sudden spikes in visual complexity, process multi-camera feeds concurrently, or handle high-frequency 60 FPS input streams without accumulating frame backlogs or dropping critical temporal data. The low latency per frame on GPU hardware confirms that spatial convolutions and feature extraction layers in YOLOv11 run with near-optimal efficiency when paired with parallel compute hardware.

Conversely, on CPU-only hardware, the pipeline achieved a throughput of 16.8 FPS. While falling below the 30 FPS threshold for full real-time video processing, this metric represents an impressive upper bound for non-accelerated deep learning workloads. The benchmark demonstrates the crucial architectural advantage of decoupling stream ingestion from inference: by offloading video demuxing, temporal slicing, and stream polling to the compiled Go application, the host system completely eliminated I/O-induced CPU stalls. In traditional single-threaded Python implementations, CPU cores frequently sit idle waiting for video frames to decode from the stream or disk. In this decoupled design, Go goroutines handle all low-level multimedia I/O, allowing the OS scheduler to dedicate maximum processor threads and vector registers (such as AVX-512 instruction sets) directly to PyTorch's heavy linear algebra routines and BLAS matrix multiplications.

Furthermore, the memory profiling metrics confirm that the contiguous array buffer implementation successfully stabilized and bounded primary RAM utilization throughout the benchmarking process. In conventional streaming applications, dynamic lists and unconstrained frame queues cause memory consumption to balloon unpredictably over time, triggering frequent operating system page faults, aggressive Python garbage collection pauses, and eventually system crashes due to memory exhaustion.

By pre-allocating a fixed 100-frame circular ring buffer in contiguous physical memory, the system established an unyielding ceiling on RAM consumption, keeping host memory overhead flat regardless of runtime duration. This predictable, low-overhead memory footprint directly demonstrates the viability of deploying this architecture on resource-constrained edge machines, such as low-power single-board computers or

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gateway devices equipped with only 8 GB of RAM. The architecture successfully reconciles high-performance video analytics with tight hardware constraints, proving that targeted software optimization can make advanced deep learning models practical for lightweight edge deployments.

CHAPTER SIX: Critical Evaluation

6.1 Strengths of the Proposed Approach

The primary strength of this architectural design lies in its strict modularity and decoupled component layout. By isolating video stream parsing, demuxing, and temporal frame extraction into an asynchronous, high-speed Go routine, the system effectively shields the inference engine from the unpredictability of video I/O. Paired with a strictly enforced 100-frame circular buffer, this decoupled design achieves an extraordinarily stable memory footprint. Because the memory required for the buffer is bounded and allocated upfront, the runtime environment completely avoids dynamic heap re-allocations, memory fragmentation, and unpredictable garbage collection pauses that typically degrade long-running Python vision pipelines.

Furthermore, processing frames as uncompressed PNG files ensures that no lossy compression artifacts such as discrete cosine transform ringing, macroblocking, or chroma subsampling errors are introduced into the pixel data. Preserving bit-perfect visual fidelity allows the advanced spatial convolutional layers and feature extraction backbones of YOLOv11 to evaluate unadulterated edge gradients and fine-grained facial geometry, maximizing detection confidence and minimizing false negatives.

6.2 System Limitations

Despite its clear operational strengths, the current architecture exhibits a significant physical trade-off in its storage workflow. Writing lossless, uncompressed PNG files directly to the physical `/dist` directory introduces continuous, high-volume disk I/O operations. Because high-definition 1080p frames possess substantial file sizes, writing dozens of uncompressed images per second generates massive data throughput across the storage bus. Over prolonged operational runtimes, this constant disk activity can easily saturate I/O queue depths, creating an artificial storage bottleneck that throttles overall frame extraction speeds.

Furthermore, when deployed on physical media particularly solid-state drives (SSDs) or flash-based embedded storage cards common in edge hardware this continuous high-throughput writing severely accelerates drive wear. Constantly overwriting flash cells leads to rapid exhaustion of the drive's Terabytes Written (TBW) endurance rating, increasing the risk of premature hardware failure and physical media degradation over time. In production environments, this limitation can be mitigated by transitioning from physical disk writes to volatile in-memory virtual filesystems (such

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as tmpfs RAM disks) or direct POSIX shared memory pointers, completely bypassing physical storage media.

6.3 Technical Challenges & Optimization

From an engineering perspective, one of the most persistent technical challenges encountered during development was reconciling the structural and dimensional disparities between classical computer vision tools and modern deep learning frameworks. OpenCV natively parses image data using a Blue-Green-Red (BGR) color model formatted as a $Height \times Width \times Channels$ ($H \times W \times C$) contiguous matrix. Conversely, PyTorch's neural network layers mandate a Red-Green-Blue (RGB) color ordering structured as a planar $Channels \times Height \times Width$ ($C \times H \times W$) tensor format. Executing this multi-step transformation—flipping color channels while simultaneously transposing array axes—initially introduced noticeable micro-stutters into the processing pipeline. These micro-stutters were caused by naive memory copying operations that fragmented the underlying memory layout. When an array is transposed without re-indexing its physical layout, it becomes non-contiguous in RAM, forcing PyTorch to perform implicit, expensive memory re-allocations on the host CPU before transferring the tensor across the PCIe bus to the GPU. To eliminate these latency spikes and restore deterministic frame delivery, the pre-processing pipeline was overhauled using low-level, vectorized NumPy primitives. By utilizing strided view manipulation combined with explicit C-contiguity enforcement via `np.ascontiguousarray()`, the channel reversal and spatial axis transposition were fused into a unified, highly optimized memory operation. This optimization ensured that the transformed arrays remained perfectly aligned in physical host memory, enabling seamless Direct Memory Access (DMA) transfers straight into GPU VRAM and permanently resolving the pre-inference micro-stutters.

CHAPTER SEVEN: Conclusion And Recommendations

7.1 Conclusion

This assignment successfully developed a highly optimized real-time facial detection inference pipeline. By integrating Go-based FFMPEG frame extraction with Python-based YOLOv11 execution, the system effectively mitigates the common memory bottlenecks associated with continuous video ingestion.

7.2 Achievement of Objectives

Table 7.1. Achievement of Objectives

Objective	Achievement	Evidence
1. Examine CNN and YOLO theoretical principles	Achieved	Chapter 2 & 3
2. Develop Go/FFMPEG extraction module	Achieved	Chapter 4
3. Implement tensor preprocessing via OpenCV	Achieved	Chapter 4
4. Conduct performance simulations	Achieved	Chapter 5
5. Analyze viability for edge hardware	Achieved	Chapter 6

7.3 Recommendations

To address the physical storage bottlenecks and drive degradation inherent to disk-based frame extraction, future iterations of this architecture should transition to a purely in-memory Inter-Process Communication (IPC) paradigm. By bypassing physical storage media entirely, the system can eliminate disk I/O latency, reduce CPU overhead associated with file system calls, and protect edge hardware from SSD write-endurance exhaustion. Rather than encoding intermediate frames into PNG files and saving them to disk, the Go parsing process and the Python inference engine can exchange raw, uncompressed RGBA pixel matrices directly within RAM using high-throughput IPC mechanisms such as POSIX shared memory segments or UNIX named pipes.

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The primary and most performant strategy involves utilizing POSIX shared memory (`shm_open` or Linux `/dev/shm`). In this framework, the Go extractor and Python consumer map the exact same physical RAM address space into their respective virtual memory regions via memory-mapped files (`mmap`). As the Go routine demuxes and decodes a video frame using FFmpeg, it writes the raw, uncompressed RGBA byte array directly into a pre-allocated ring buffer residing in shared memory. Python can then instantiate a zero-copy NumPy array directly over those memory addresses using `numpy.frombuffer()` or Ctypes pointers. Because this approach transfers pointers rather than copying data blocks across memory spaces, frame transfer latency drops from milliseconds to sub-microseconds. Transfer speeds become bounded only by system RAM bus bandwidth (operating at tens of gigabytes per second), ensuring that the ingestion pipeline operates at near-zero communication cost.

To maintain thread safety and prevent race conditions within shared memory such as Python reading a frame while Go is actively writing over it, the architecture must implement low-level synchronization primitives. Lightweight inter-process mutexes or POSIX semaphores can be managed across the Go-Python boundary to coordinate read/write offsets in the circular buffer. Alternatively, UNIX named pipes (FIFOs) or UNIX domain sockets offer a robust stream-oriented alternative. While named pipes involve a minor memory copy through kernel buffer space, they natively handle process synchronization and non-blocking stream execution, allowing Python to continuously read raw RGBA byte streams as a contiguous sequence of fixed-size byte arrays without needing complex semaphore management.

Crucially, streaming raw RGBA matrices directly between processes completely eliminates the CPU cycles previously spent on PNG encoding in Go and PNG decoding in OpenCV. A raw 1080p RGBA frame consists of exactly $1920 \times 1080 \times 4 = 8,294,400$ bytes; because this dimensional footprint is mathematically constant, both processes can manipulate the buffer using fixed byte offsets. Python can perform in-place array transpositions directly on the mapped buffer, preparing the tensor for YOLOv11 CUDA ingestion with zero intermediate allocations. Bypassing disk I/O entirely transforms the ingestion pipeline into a closed, high-efficiency memory loop, significantly boosting end-to-end FPS, ensuring true deterministic throughput, and making 24/7 continuous stream analytics feasible on lightweight edge devices.

7.4 Future Work

Future investigations could explore: Integrating facial tracking algorithms (e.g., DeepSORT) to maintain individual identities across frames. Fine-tuning the YOLOv11 model To address the physical storage bottlenecks and drive degradation inherent to disk-based frame extraction, future iterations of this architecture should transition to a purely in-memory Inter-Process Communication (IPC) paradigm. By bypassing physical storage media entirely, the system can eliminate disk I/O latency, reduce CPU overhead associated with file system calls, and protect edge hardware from SSD write-endurance exhaustion. Rather than encoding intermediate frames into PNG files and saving them to disk, the Go parsing process and the Python inference engine can exchange raw, uncompressed RGBA pixel matrices directly within RAM using high-throughput IPC mechanisms such as POSIX shared memory segments or UNIX named pipes.

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BIOGRAPHIES



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a very passionate developer, passionate about performant and beautiful code. :-)

he is the reason GO was involved, also helped a ton n creating this IEEE style paper



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